

Idris Abdulmumin, PhD

African NLP • Machine Translation • Speech & Language Resources

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Research Profile

Postdoctoral researcher and educator specialising in Natural Language Processing for African and other low-resource languages. My work spans machine translation, automatic speech recognition, and the analysis of sentiment, emotion, hate, and offensive language, with a strong emphasis on building openly available datasets, benchmarks, and models that serve under-represented communities. I lead and co-organise large multilingual research efforts across SemEval, WMT, IWSLT, and AfricaNLP, and supervise postgraduate research at several African institutions.

Education

- Jan. 2023 📖 **Ph.D. Computer Science, Bayero University, Kano.**
Thesis title: *Domain-Aware Self-Learning and Synthetic Data Transfer for Efficient Back-Translation in Low Resource Neural Machine Translation.*
- Jan. 2017 📖 **M.Sc. IT Security, Nottingham Trent University, Nottingham, UK.**
Dissertation title: *Web Application Security Analysis.*
- Apr. 2013 📖 **B.Sc. Computer Science, Bayero University, Kano.**
Project title: *Plagiarism Checker, File Comparer and Project Library Software.*

Employment

- 2025, 2026 📖 **Visiting Professor,**
African Institute for Mathematical Sciences (AIMS) – South Africa.
- 2023 – 2026 📖 **Postdoctoral Fellow,**
Data Science for Social Impact Research Group, University of Pretoria – South Africa.
- 2023 📖 **Lecturer I,**
Department of Computer Science, Ahmadu Bello University, Zaria, Kaduna–Nigeria.
- 2020 – 2023 📖 **Lecturer II,**
Department of Computer Science, Ahmadu Bello University, Zaria, Kaduna–Nigeria.
- 2017 – 2020 📖 **Assistant Lecturer,**
Department of Computer Science, Ahmadu Bello University, Zaria, Kaduna–Nigeria.

Teaching

- 2025 📖 **African Institute for Mathematical Sciences, South Africa**
Natural Language Processing and LLMs (MSc)
- 📖 **University of Pretoria**
COS760: Natural Language Processing (Honors)

Teaching (continued)

- 2023 – ...  **Arewa Data Science Academy**
Natural Language Processing
- 2017 – 2023  **Ahmadu Bello University, Zaria**
COSC211: Object-Oriented Programming I
COSC212: Object-Oriented Programming II
COSC406: Advanced Database Systems
COSC403: Software Engineering
COSC802: Software Engineering Methodologies (MSc)

Supervision

University of Pretoria

PhD

- **Mokgadi Penelope Matloga** (2025 – . . .): Investigating Efficient Few Shot Learning Approaches for Natural Language Classification Tasks

Masters, Big Data Science

- **Rozina Myoya** (2024): Analysing Public Transport User Sentiment
- **Adinda Behrens** (2025): Bridging the Economic Data Divide: Leveraging Natural Language Processing and Media Sources to Improve Statistical Coverage in Africa
- **Reshay Mahepal** (2025): Evaluating the Efficacy of Zero-Shot and Few-Shot Prompting for Customer Intent Classification in the Banking Sector Using Large Language Models
- **Juliet Rushambwe** (2026): Active Learning for Data Augmentation in Low Resource African Neural Machine Translation
- **Abiola Yetunde Akinbowale** (2026): Harmonizing Multilingual News Categories: A Unified Approach to African Language News Classification
- **Realeboga Monare** (2026): User-Centric Evaluation of Large Language Models on Sepedi versus English
- **Akhona Stafane** (Ongoing): Development of a GPT-based Transformer Model for Customer Analysis and Next Transaction Prediction in the Banking Sector

Honors

- **Kgabo Komane** (2024): Sentiment Analysis with Sarcasm Detection in Setswana Language
- **Zizo Majiza** (2024): Summarization of Supreme Court of Appeals judgements using deep learning techniques
- **Zainab Abdulrasaq** (Ongoing): AfriGemma: A Multimodal, Multilingual Pan-African Large Language Model
- **Andinda Bakainaga** (Ongoing): AfriGemma: A Multimodal, Multilingual Pan-African Large Language Model

Supervision (continued)

Ahmadu Bello University, Zaria

■ BSc

- **16 Students** (2017 – 2023): Supervised various final year projects in Computer Science and Software Engineering.

Selected Research Publications

Journal Articles

- 1 **Abdulmumin, I.**, & Marivate, V. (2025). Zasca-sum: A dataset of the south africa supreme courts of appeal judgments and media summaries for legal documents summarization research. *Data in Brief*, 111567. [doi:https://doi.org/10.1016/j.dib.2025.111567](https://doi.org/10.1016/j.dib.2025.111567)
- 2 **Abdulmumin, I.**, Galadanci, B. S., & Aliyu, G. (2021). Tag-less back-translation. *Machine Translation*, 35(4), 519–549. [doi:10.1007/s10590-021-09284-y](https://doi.org/10.1007/s10590-021-09284-y)

Conference Proceedings

- 1 Imam, S. H., Bello, M. Y., Umar, H. A., Belay, T. D., **Abdulmumin, I.**, Yimam, S. M., & Muhammad, S. H. (2026, March). Full fine-tuning vs. parameter-efficient adaptation for low-resource African ASR: A controlled study with whisper-small. In E. A. Chimoto, C. Lignos, S. Muhammad, **I. Abdulmumin**, C. Siro, & D. I. Adelani (Eds.), *Proceedings of the 7th workshop on African natural language processing (AfricaNLP 2026)* (pp. 197–203). Rabat, Morocco: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2026.africanlp-main.19/>
- 2 Belay, T. D., Ademtew, H. B., **Abdulmumin, I.**, Imam, S. H., Chilala, A. J., Agyapong, G., ... Muhammad, S. H. (2026). Trust but check: LLM-assisted review of human translations in african languages. In *7th workshop on african natural language processing*. Retrieved from <https://openreview.net/forum?id=8B2WDhIAMV>
- 3 Belay, T. D., Hussen, K. Y., Imam, S. H., Ahmad, I. S., Inuwa-Dutse, I., Aabr, ... Muhammad, S. H. (2026). The rise of africaNLP: Contributions, contributors, and community impact (2005–2025). In *7th workshop on african natural language processing*. Retrieved from <https://openreview.net/forum?id=DWCxhpad1k>
- 4 Belay, T. D., Azime, I. A., Ahmad, I. S., Adelani, D. I., **Abdulmumin, I.**, Ayele, A. A., ... Yimam, S. M. (2025, November). AfroXLMR-social: Adapting pre-trained language models for African languages social media text. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Findings of the association for computational linguistics: Emnlp 2025* (pp. 15570–15587). [doi:10.18653/v1/2025.findings-emnlp.842](https://doi.org/10.18653/v1/2025.findings-emnlp.842)
- 5 Dale, D., Burchell, L., Maillard, J., **Abdulmumin, I.**, Anastasopoulos, A., Caswell, I., & Koehn, P. (2025, November). Findings of the WMT 2025 shared task of the open language data initiative. In B. Haddow, T. Kocmi, P. Koehn, & C. Monz (Eds.), *Proceedings of the tenth conference on machine translation* (pp. 495–502). [doi:10.18653/v1/2025.wmt-1.26](https://doi.org/10.18653/v1/2025.wmt-1.26)
- 6 **Abdulmumin, I.**, Agostinelli, V., Alumäe, T., Anastasopoulos, A., Bentivogli, L., Bojar, O., ... Züfle, M. (2025, July). Findings of the IWSLT 2025 evaluation campaign. In E. Salesky, M. Federico, & A. Anastasopoulos (Eds.), *Proceedings of the 22nd international conference on spoken language translation (iwslt 2025)* (pp. 412–481). [doi:10.18653/v1/2025.iwslt-1.44](https://doi.org/10.18653/v1/2025.iwslt-1.44)

- 7 Imam, S. H., Sani, B., Gete, D. K., Ahmed, B. Y., Ahmad, I. S., **Abdulumumin, I.**, ... Muhammad, S. H. (2025, July). Automatic speech recognition for African low-resource languages: Challenges and future directions. In C. Lignos, **I. Abdulumumin**, & D. Adelani (Eds.), *Proceedings of the sixth workshop on african natural language processing (africanlp 2025)* (pp. 89–94). [doi:10.18653/v1/2025.africanlp-1.13](https://doi.org/10.18653/v1/2025.africanlp-1.13)
- 8 Muhammad, S. H., Ousidhoum, N., **Abdulumumin, I.**, Wahle, J. P., Ruas, T., Beloucif, M., ... Mohammad, S. M. (2025, July). BRIGHTER: BRIdging the gap in human-annotated textual emotion recognition datasets for 28 languages. In W. Che, J. Nabende, E. Shutova, & M. T. Pilehvar (Eds.), *Proceedings of the 63rd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 8895–8916). [doi:10.18653/v1/2025.acl-long.436](https://doi.org/10.18653/v1/2025.acl-long.436)
- 9 Muhammad, S. H., Ousidhoum, N., **Abdulumumin, I.**, Yimam, S. M., Wahle, J. P., Lima Ruas, T., ... Mohammad, S. (2025, July). SemEval-2025 task 11: Bridging the gap in text-based emotion detection. In S. Rosenthal, A. Rosá, D. Ghosh, & M. Zampieri (Eds.), *Proceedings of the 19th international workshop on semantic evaluation (semeval-2025)* (pp. 2558–2569). Vienna, Austria: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2025.semeval-1.327/>
- 10 Ortega, J. E., Joel Zevallos, R., Chen, W., & **Abdulumumin, I.** (2025, July). QUESPA submission for the IWSLT 2025 dialectal and low-resource speech translation task. In E. Salesky, M. Federico, & A. Anastasopoulos (Eds.), *Proceedings of the 22nd international conference on spoken language translation (iwslt 2025)* (pp. 260–268). [doi:10.18653/v1/2025.iwslt-1.25](https://doi.org/10.18653/v1/2025.iwslt-1.25)
- 11 **Abdulumumin, I.**, Mkhwanazi, S., Mbooi, M., Muhammad, S. H., Ahmad, I. S., Putini, N., ... Mari-vate, V. (2024, November). Correcting FLORES evaluation dataset for four African languages. In B. Haddow, T. Kocmi, P. Koehn, & C. Monz (Eds.), *Proceedings of the ninth conference on machine translation* (pp. 570–578). [doi:10.18653/v1/2024.wmt-1.44](https://doi.org/10.18653/v1/2024.wmt-1.44)
- 12 Ousidhoum, N., Muhammad, S., Abdalla, M., **Abdulumumin, I.**, Ahmad, I., Ahuja, S., ... Mohammad, S. (2024, August). SemRel2024: A collection of semantic textual relatedness datasets for 13 languages. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the association for computational linguistics: Acl 2024* (pp. 2512–2530). [doi:10.18653/v1/2024.findings-acl.147](https://doi.org/10.18653/v1/2024.findings-acl.147)
- 13 Muhammad, S., **Abdulumumin, I.**, Ayele, A., Ousidhoum, N., Adelani, D., Yimam, S., ... Arthur, S. (2023, December). AfriSenti: A Twitter sentiment analysis benchmark for African languages. In H. Bouamor, J. Pino, & K. Bali (Eds.), *Proceedings of the 2023 conference on empirical methods in natural language processing* (pp. 13968–13981). Singapore: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2023.emnlp-main.862>
- 14 **Abdulumumin, I.**, Khalid, A. A., Muhammad, S. H., Ahmad, I. S., Aliyu, L. J., Sani, B., ... Hassan, S. A. (2023, November). Leveraging closed-access multilingual embedding for automatic sentence alignment in low resource languages. In *Proceedings of the international conference on computing and advances in information technology (ICCAIT 2023)*, Zaria, Nigeria. Retrieved from <https://arxiv.org/abs/2311.12179>
- 15 Adelani, D. I., Masiak, M., Azime, I. A., Alabi, J., Tonja, A. L., Mwase, C., ... Stenetorp, P. (2023, November). Masakhanews: News topic classification for african languages. In *Proceedings of the 13th international joint conference on natural language processing and the 3rd conference of the asia-pacific chapter of the association for computational linguistics* (pp. 144–159). Nusa Dua, Bali: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2023.ijcnlp-long.10>
- 16 Muhammad, S. H., **Abdulumumin, I.**, Yimam, S. M., Adelani, D. I., Ahmad, I. S., Ousidhoum, N., ... Ruder, S. (2023, July). SemEval-2023 task 12: Sentiment analysis for African languages

(AfriSenti-SemEval). In *Proceedings of the the 17th international workshop on semantic evaluation (semeval-2023)* (pp. 2319–2337). Toronto, Canada: Association for Computational Linguistics. Retrieved from <https://aclanthology.org/2023.semeval-1.315>

- 17 **Abdulmumin, I.**, Beukman, M., Alabi, J., Emezue, C. C., Chimoto, E., Adewumi, T., ... Gwadabe, T. (2022, December). Separating grains from the chaff: Using data filtering to improve multilingual translation for low-resourced african languages. In *Proceedings of the seventh conference on machine translation* (pp. 1001–1014). Abu Dhabi: Association for Computational Linguistics.
- 18 **Abdulmumin, I.**, Dash, S. R., Dawud, M. A., Parida, S., Muhammad, S., Ahmad, I. S., ... Bello, B. S. (2022, June). Hausa visual genome: A dataset for multi-modal english to hausa machine translation. In *Proceedings of the language resources and evaluation conference* (pp. 6471–6479). Marseille, France: European Language Resources Association. Retrieved from <https://aclanthology.org/2022.lrec-1.694>

Book Chapters

- 1 **Abdulmumin, I.**, Galadanci, B. S., Ahmad, I. S., & Abdullahi, R. I. (2021). Data selection as an alternative to quality estimation in self-learning for low resource neural machine translation. In O. Gervasi, B. Murgante, S. Misra, C. Garau, I. Blečić, D. Taniar, ... C. M. Torre (Eds.), *Computational science and its applications – iccsa 2021* (pp. 311–326). [doi:10.1007/978-3-030-87013-3_24](https://doi.org/10.1007/978-3-030-87013-3_24)
- 2 **Abdulmumin, I.**, Galadanci, B. S., & Isa, A. (2021). Enhanced back-translation for low resource neural machine translation using self-training. In S. Misra & B. Muhammad-Bello (Eds.), *Information and communication technology and applications* (pp. 355–371). [doi:10.1007/978-3-030-69143-1_28](https://doi.org/10.1007/978-3-030-69143-1_28)

Preprints

- 1 **Abdulmumin, I.**, Gwadabe, T., Muhammad, S. H., Adelani, D. I., Khalo, N., Ahmad, I. S., ... Marivate, V. (2026). Afriscience-mt: Towards decolonizing science in africa through text translation. arXiv: 2605.29741 [cs.CL]. Retrieved from <https://arxiv.org/abs/2605.29741>
- 2 **Abdulmumin, I.**, Matloga, M. P., Belay, T. D., Kondowe, B., Mohleleng, L., Letsoalo, H. N., ... Marivate, V. (2026). Temporal simultaneity predicts annotation quality in sentiment corpora. arXiv: 2605.27239 [cs.CL]. Retrieved from <https://arxiv.org/abs/2605.27239>
- 3 Becker, J., Yu, L.-C., Muhammad, S. H., Wahle, J. P., Ruas, T., **Abdulmumin, I.**, ... Mohammad, S. M. (2026). Dimstance: Multilingual datasets for dimensional stance analysis. arXiv: 2601.21483 [cs.CL]. Retrieved from <https://arxiv.org/abs/2601.21483>
- 4 Marivate, V., Olaleye, K., Mundia, S., Bakainga, A., Netshifhefhe, U., Milanzie, M., ... Rananga, S. (2026). Swivuriso: The south african next voices multilingual speech dataset. arXiv: 2512.02201 [cs.CL]. Retrieved from <https://arxiv.org/abs/2512.02201>
- 5 Suarez, P. O., Burchell, L., Arnett, C., Mosquera-Gómez, R., Hincapie-Monsalve, S., Vaughan, T., ... Luger, S. (2026). Commonlid: Re-evaluating state-of-the-art language identification performance on web data. arXiv: 2601.18026 [cs.CL]. Retrieved from <https://arxiv.org/abs/2601.18026>
- 6 Tonja, A. L., Anand, S., Villa-Cueva, E., Azime, I. A., Alabi, J. O., Mohamed, M. A., ... Solorio, T. (2026). Afri-mcqa: Multimodal cultural question answering for african languages. arXiv: 2601.05699 [cs.CL]. Retrieved from <https://arxiv.org/abs/2601.05699>

Experience and Portfolio

Teamwork and Collaboration with Others

- 2025
- **Technical Lead:** Developed a Telegram Bot for the collection of over 1,000 hours of ASR data in the Africa Next Voices Project.
 - **Technical Lead:** Developed a web-based Annotation Tool for the collection of Aspect-based Sentiment Analysis dataset in 5 African Languages.
 - **Co-organizer:** WMT Shared Task, title: "[Open Language Data Initiative \(OLDI\)](#)".
 - **Co-organizer:** SemEval 2026 Detecting Multilingual, Multicultural and Multi-event Online Polarization.
 - **Co-organizer:** SemEval 2026 Dimensional Aspect-Based Sentiment Analysis and Stance Detection.
- 2024
- **Co-organizer:** SemEval 2025 Task 11, title: "Bridging the Gap in Text-Based Emotion Detection", including 17 African languages.
 - **Co-organizer:** English-to-Lowres Multi-Modal Translation Task at the WMT 2024.
- 2023
- **Co-organizer:** SemEval 2024 Task 1, title: "Semantic Textual Relatedness for African and Asian Low Resource Languages".
 - **Co-organizer:** Large Scale Multilingual African Track at the WMT 2023.
 - **Experimental Lead:** Machine Translation, Lacuna Funded Decolonise Science Project.
- 2022
- **Co-organizer:** "Bird of the Feather Workshop—Challenges for Hate Speech Detection in Low-resource African languages and Beyond at EMNLP 2022".
 - **Co-organizer & Technical Lead:** SemEval 2023 Task 12, title: "AfriSenti-SemEval: Sentiment Analysis for Low-resource African Languages using Twitter Dataset".
 - **Co-investigator & Project Manager:** Creation of language resource, title: "AfriHate: Hate and Offensive Speech Dataset for African Languages". (Lacuna 2021 Grant)
 - **Collaborator & Technical Lead:** Ndyuka Translate - Automatic Translation System for Ndyuka, Pamaka & Aluku to/from English and French.
 - **Researcher & Collaborator:** [MasakhaNER 2.0](#): Curating named-entity recognition datasets for African languages.
 - **Researcher & Collaborator:** [LAFAND-MT](#): Curating machine translation datasets for African languages.
- 2021
- **Project Lead:** Hausa Visual Genome: A Dataset for Multi-Modal English to Hausa Machine Translation (completed, [published](#)).
 - **Project Manager:** Corpus, Sentiment, and Hate Speech Lexicon for Major Nigerian Languages, Lacuna Language 2020 Funded Project (completed, [published](#)).

Professional Affiliation

- 2022 – ...
- **Member,** [European COST Action LITHME](#) (Language in the Human-Machine Era)

Experience and Portfolio (continued)

Awards and Achievements

- 2025  Principal Investigator: AfriGemma: Advancing African Language NLP for Research, Healthcare, and Societal Benefit.
- 2022  Co-Investigator: Artificial Intelligence for Education Innovation Research Network in Africa.
-  Co-Investigator: Google CSR Award.
-  Co-Investigator: Lacuna Fund 2021 Language Datasets Award.
-  Co-Investigator: Oracle OCI Award.
- 2021  Co-Investigator: Lacuna Fund 2020 Language Datasets Award.

Skills

- Core  **Consultation:** Provide expert advice in African Natural Language Processing (e.g., Machine Translation, Automatic Speech Recognition (ASR) and Text-to-Speech (TTS), Sentiment, Emotion, Hate, and Offensive Language Analysis).
Academic research: Conduct interdisciplinary research and publish peer-reviewed papers.
Teaching, supervision, and training: Deliver lectures, workshops, mentoring sessions, and research supervision to undergraduate and postgraduate students.
- Languages  **Hausa:** Native.
English: Excellent reading, writing, and speaking proficiency.
Nigerian Pidgin: Proficient in reading, writing, and speaking.
- Coding  Python, Java, php, C++, sql, $\LaTeX$.
- Databases  Mysql.

References

Available on Request